Formal Proof of Scaling Parameter Invariance

Problem Statement

For a copula C on $[0,1]^2$, direction parameter $\delta \in (0,2)$, and scaling parameter $\gamma > 0$, the generalized lower-tail dependence coefficient is defined as

$$\kappa_L(\delta, \gamma; C) = \lim_{u \downarrow 0} \frac{\log C\big(\gamma u^{\delta}, \gamma^{-1} u^{2-\delta}\big)}{\log u},$$

whenever the limit exists. We aim to rigorously prove that under suitable regularity conditions, the value of $\kappa_L(\delta, \gamma; C)$ is independent of the choice of γ. Furthermore, we demonstrate that the proposed Pareto-mixture copula C_{PM} satisfies this claim.

Regularity Assumptions

To formalize the analysis of the limit, let us fix a reference scaling parameter $\gamma^* > 0$. We define the functional mapping for the pre-limit quantity as:

$$\Lambda(u, d, \gamma) := \frac{\log C\big(\gamma u^{d}, \gamma^{-1} u^{2-d}\big)}{\log u}, \quad \text{for } u \in (0,1).$$

We operate under the following standard regularity conditions concerning the tail behavior of the copula:

- **A1 (Existence and Finiteness):** For the given $\delta \in (0,2)$, the limit $\lambda(d) := \kappa_L(d, \gamma^*; C) = \lim_{u \downarrow 0} \Lambda(u, d, \gamma^*)$ exists and is finite for all d in a neighborhood of δ.
- **A2 (Local Uniform Convergence):** The convergence to the limit function $\lambda(\cdot)$ is locally uniform at δ. That is, for every $\varepsilon > 0$, there exist a radius $\nu > 0$ and a threshold $u_0 \in (0,1)$ such that for all $u \in (0, u_0]$ and all d satisfying $|d - \delta| \le \nu$,

$$|\Lambda(u, d, \gamma^*) - \lambda(d)| \le \varepsilon.$$

- **A3 (Continuity):** The limit function $d \mapsto \lambda(d)$ is continuous at δ.

Proof of Invariance

Let $\gamma > 0$ be an arbitrary alternative scaling parameter. We must prove that $\lim_{u \downarrow 0} \Lambda(u, \delta, \gamma) = \lambda(\delta)$.

Step 1: Establishing the Shift Identity. We first relate the evaluation of Λ at γ to its evaluation at the reference γ^* by introducing a u-dependent shift in the direction parameter. Let $\rho = \log(\gamma/\gamma^*)$. For any $u \in (0,1)$, note that $\log u < 0$. We define a perturbed direction parameter:

$$\delta(u) := \delta + \frac{\rho}{\log u}.$$

Utilizing the properties of logarithms and exponentials (specifically $e^{\log u} = u$ and $e^{\rho} = \gamma/\gamma^*$), we can rewrite the arguments of the copula:

$$\gamma^* u^{\delta(u)} = \gamma^* \exp\left\{ \left(\delta + \frac{\rho}{\log u} \right) \log u \right\} = \gamma^* e^{\delta \log u} e^{\rho} = \gamma u^{\delta},$$
$$(\gamma^*)^{-1} u^{2-\delta(u)} = (\gamma^*)^{-1} \exp\left\{ \left(2 - \delta - \frac{\rho}{\log u} \right) \log u \right\} = (\gamma^*)^{-1} e^{(2-\delta)\log u} e^{-\rho} = \gamma^{-1} u^{2-\delta}.$$

Consequently, substituting these equivalent arguments into our functional mapping yields the exact identity:

$$\Lambda(u, \delta, \gamma) = \Lambda(u, \delta(u), \gamma^*), \quad \text{for all } u \in (0,1). \tag{1}$$

Step 2: The ε-Argument. We proceed to show that $\lim_{u\downarrow 0} |\Lambda(u,\delta,\gamma) - \Lambda(u,\delta,\gamma^*)| = 0$. Fix an arbitrary $\varepsilon > 0$. Using the identity in (1), we apply the triangle inequality by adding and subtracting the theoretical limit values evaluated at both the perturbed and reference directions, $\lambda(\delta(u))$ and $\lambda(\delta)$:

$$\begin{aligned} |\Lambda(u,\delta,\gamma) - \Lambda(u,\delta,\gamma^*)| &= |\Lambda(u,\delta(u),\gamma^*) - \Lambda(u,\delta,\gamma^*)| \\ &\le \underbrace{|\Lambda(u,\delta(u),\gamma^*) - \lambda(\delta(u))|}_{\text{Term I}} + \underbrace{|\lambda(\delta(u)) - \lambda(\delta)|}_{\text{Term II}} + \underbrace{|\lambda(\delta) - \Lambda(u,\delta,\gamma^*)|}_{\text{Term III}}. \end{aligned} \tag{2}$$

We now bound each term on the right-hand side of (2) by $\varepsilon/3$ for sufficiently small u:

1. **Bounding Term III:** By the fundamental definition of the limit (Assumption A1), there exists $u_1 \in (0,1)$ such that for all $u \in (0,u_1]$,
$$|\lambda(\delta) - \Lambda(u,\delta,\gamma^*)| \le \frac{\varepsilon}{3}.$$
2. **Bounding Term II:** Observe that as $u \downarrow 0$, $\log u \to -\infty$. Since ρ is a fixed constant, $\lim_{u\downarrow 0} \delta(u) = \delta$. Because $\lambda(\cdot)$ is continuous at δ (Assumption A3), there exists $u_2 \in (0,1)$ such that for all $u \in (0,u_2]$,
$$|\lambda(\delta(u)) - \lambda(\delta)| \le \frac{\varepsilon}{3}.$$
3. **Bounding Term I:** By the locally uniform convergence condition (Assumption A2), there exists a radius $\nu > 0$ and a threshold $u_0 \in (0,1)$ bounding the convergence error to $\varepsilon/3$ for all $d \in [\delta-\nu, \delta+\nu]$. Since $\delta(u) \to \delta$, there exists $u_3 \in (0,1)$ such that $|\delta(u) - \delta| \le \nu$ for all $u \in (0,u_3]$. Let $u_4 = \min(u_0,u_3)$. For all $u \in (0,u_4]$, the perturbed direction $\delta(u)$ lies within the uniform convergence window, ensuring:
$$|\Lambda(u,\delta(u),\gamma^*) - \lambda(\delta(u))| \le \frac{\varepsilon}{3}.$$

Step 3: Conclusion of the Limit. Let $u^* = \min(u_1,u_2,u_4)$. For all $u \in (0,u^*]$, substituting the three bounds back into (2) yields:

$$|\Lambda(u,\delta,\gamma) - \Lambda(u,\delta,\gamma^*)| \le \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Because $\varepsilon > 0$ was arbitrary, it strictly follows that $\lim_{u\downarrow 0} \Lambda(u,\delta,\gamma) = \lim_{u\downarrow 0} \Lambda(u,\delta,\gamma^*) = \lambda(\delta)$.

We have shown $\kappa_L(\delta,\gamma;C) = \kappa_L(\delta,\gamma^*;C)$. As both $\delta \in (0,2)$ and $\gamma > 0$ were arbitrarily chosen, the generalized lower-tail dependence coefficient is invariant to the choice of the scaling parameter γ. □

Remark 1. *If the lower-tail dependence is infinite, i.e., $\lambda(\delta) = +\infty$, the proof framework naturally adapts using an M-bound approach. For any arbitrarily large $M > 0$, local uniformity guarantees a threshold u_0 such that $\Lambda(u,d,\gamma^*) \ge M$ for all d near δ. As $\delta(u) \to \delta$, $\Lambda(u,\delta(u),\gamma^*) \ge M$ for sufficiently small u, confirming that $\lim_{u\downarrow 0} \Lambda(u,\delta,\gamma) = +\infty$ and thus exact invariance is preserved even in the divergent case.*

Verification for the Pareto-Mixture Copula C_{PM}

The proposed Pareto-mixture copula C_{PM} defined via the independent Pareto factor random variables $(R, R^*, H_1, H_2, H_1^*, H_2^*)$ inherently satisfies the analytic regularity assumptions (A1-A3) required for this proof.

Specifically, the distribution of random variables constructed through products and quotients of Pareto-distributed components belongs to the domain of multivariate regular variation. Due to the scale-invariant properties of regular variation, the joint tail probability distribution admits a smooth, homogeneous asymptotic expansion. Consequently, the logarithmic tail ratio $\Lambda_{\mathrm{PM}}(u,d,\gamma) = \frac{\log C_{\mathrm{PM}}(\gamma u^d, \gamma^{-1} u^{2-d})}{\log u}$ evaluates asymptotically to a continuous algebraic function of the angular parameter d.

Because the tail exponent function governing C_{PM} is well-behaved and free of singularities inside the parameter space $d \in (0,2)$, the local uniform convergence (Assumption A2) and limit continuity (Assumption A3) natively hold. Thus, the general ε-proof applies identically to the proposed model: the structural dependence metric $\kappa_L(\delta,\gamma;C_{\mathrm{PM}})$ isolates the purely angular properties governed by δ, completely invariant to the baseline scale γ.